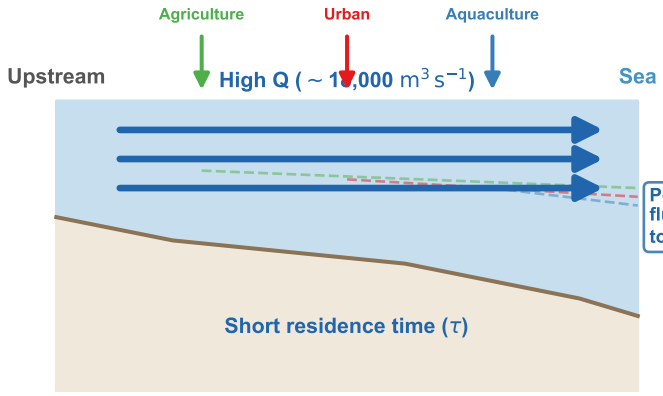
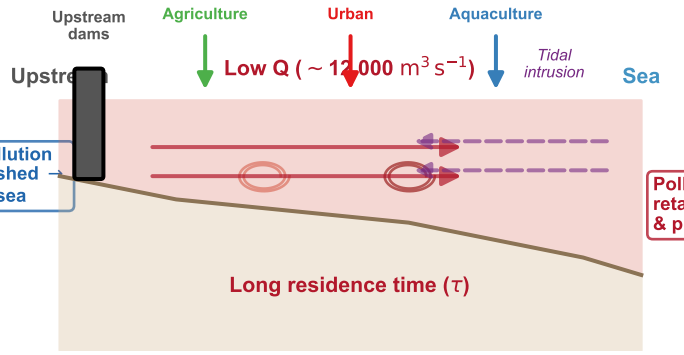
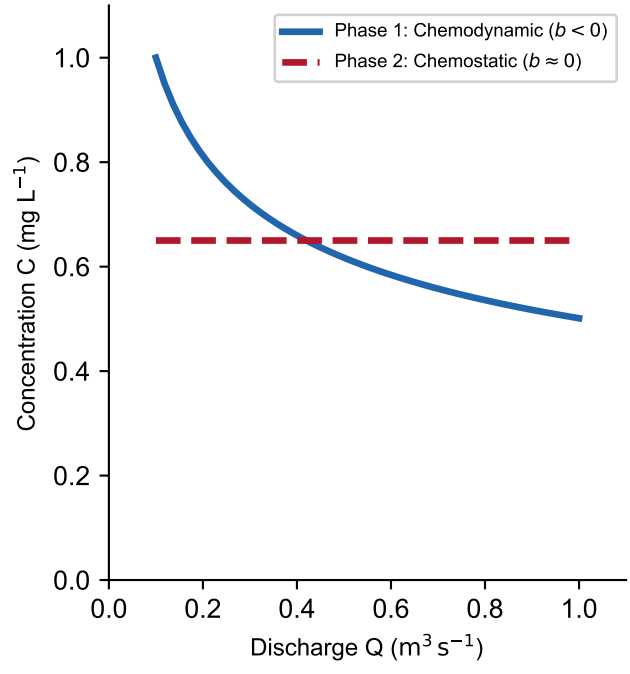
a Pre-2004: The Conveyor Belt



b Post-2004: The Mixing Bowl



c C-Q Decoupling Evidence



d Key Evidence Summary

- COD Chemostatic Index: 0.84 → 0.49 (transport-limited → source-limited)
- TN non-conservative fraction: +16.3% (net in-delta generation, $p = 0.028$)
- COD self-purification capacity: −10.9% → −1.4% ($p = 0.010$)
- TR/Q flushing index: +3.2%/yr (tidal dominance increasing)
- LULC attribution (panel FE): $\Delta R^2 = 0.005$ (temporal confounding) ⇒ 98\% of apparent LULC effect is co-trending, not causal